

PEBBLING NUMBERS OF FOUR CUBIC 12-VERTEX GRAPHS OF GIRTH 3 AND DIAMETER 3: ONE IS CLASS 0, THE OTHER THREE ARE 14

ABSTRACT. In the final remarks of *The only Class 0 Flower snark is the smallest* (arXiv:2505.22941v2), Bridi, Martins, Marquezino and de Figueiredo suggest investigating whether the pebbling number of the House of Graphs graphs #1395 (the truncated tetrahedron), #44170 and #44172 is indeed 13, and whether #6698 is Class 0. We settle both questions for the four explicit cubic 12-vertex graphs of girth 3 and diameter 3 whose adjacency lists are printed below; these lists were served by the House of Graphs REST API for those four ids, but we do not verify that identification, and every statement here is a statement about the lists as printed. For three of them the suggested value 13 does not hold: $\pi = 14$, and each excess is witnessed by an explicit 13-pebble unsolvable configuration whose unsolvability is certified by the finite move-closed set of configurations reachable from it, so that lower bound rests on the closure of three single configurations rather than on an exhaustive sweep of the configurations of a given total. The fourth has $\pi = 12$, i.e. it *is* Class 0, and it is not isomorphic to J_3 ; so the Flower snark J_3 is not the only Class 0 graph among the cubic 12-vertex graphs of girth 3 and diameter 3.

1. INTRODUCTION

We use the definitions of the source paper verbatim [1, §2]. A *configuration* on a graph G is a map $C: V(G) \rightarrow \mathbb{N}$, where $C(v)$ is the number of pebbles at v ; its *total* is $\sum_v C(v)$. A *pebbling move* removes two pebbles from a vertex v and places one pebble on a neighbour $u \in N(v)$. If some sequence of pebbling moves places a pebble on a target vertex r , then C is *r -solvable*; otherwise C is *r -unsolvable*. The *pebbling number* $\pi(G)$ is the least t such that every configuration of total t is r -solvable for every $r \in V(G)$. Always $\pi(G) \geq n(G)$, and G is *Class 0* if $\pi(G) = n(G)$.

The source paper proves that the Flower snark J_3 is Class 0, i.e. $\pi(J_3) = 12$, and closes with the following suggestion, which we quote verbatim from the final remarks of its `arXiv v2 LATEX` source:

As a suggestion of application, one natural direction is to consider other cubic graphs with 12 vertices, girth 3, and diameter 3—the same parameters as the Flower snark J_3 . According to the House of Graphs database [2], only five such graphs exist. Besides J_3 , the others include graphs with IDs #1395, #6698, #44170, and

2020 *Mathematics Subject Classification*. 05C57, 05C12, 05C30.

Key words and phrases. graph pebbling, pebbling number, Class 0 graph, cubic graph, truncated tetrahedron, flower snark, House of Graphs, exhaustive search.

#44172. The graph #1395 is known as the truncated tetrahedral graph. We find unsolvable configurations with 12 pebbles for the graphs #1395, #44170, and #44172 (see Figure [reference omitted]), implying that these graphs are not Class 0. One could investigate whether the pebbling number of these three graphs is indeed 13; and whether graph #6698 is Class 0, i.e., determine whether J_3 is the only Class 0 graph among the five with the same parameters.

Only the presentation is altered: the citation of the database is re-keyed to our bibliography (our [2] is the source's own database reference), the bracketed note stands for the source's cross-reference to its figure of 12-pebble unsolvable configurations, which we do not reproduce, and a footnote giving the database address is omitted.

The suggestion has two halves. The first offers a value tentatively rather than asserting it: the source's own figure of 12-pebble unsolvable configurations already gives $\pi \geq 13$ for the three graphs, so what its suggested value 13 would require is the upper bound $\pi \leq 13$. The second half is posed neutrally.

Throughout, #1395, #6698, #44170 and #44172 are names for the four adjacency lists printed in §2, and nothing below is a claim about the House of Graphs database. Those lists were served by the database's REST API for the four ids named in the quotation; we could not afterwards read the database's holdings, since its search endpoint returned HTTP 401 to us, so the identification of the printed lists with those ids is not verified here and is not verified by the accompanying program. Three of the lists are, however, pinned to the source's own figure of 12-pebble unsolvable configurations by an exact edge-set identity: reading the node labels of each of its three subfigures as API index = label – 1 reproduces the fetched adjacency list of the corresponding graph exactly, 18 = 18 edges with empty symmetric difference. For #6698, which the source's figure does not show, there is no such pin.

Theorem 1. $\pi(\#1395) \geq 14$, $\pi(\#44170) \geq 14$ and $\pi(\#44172) \geq 14$. *In particular the suggested value 13 does not hold for any of the three graphs.*

Theorem 1 is carried by the three configurations of Table 1, each of total 13; §3 certifies each of them by the finite set of configurations reachable from it, which is closed under pebbling moves and contains no configuration with a pebble on the target. Checking such a certificate is the deterministic closure of one configuration: still a finite search of the configurations reachable from it, but not an enumeration of the configurations of a given total. The three closures have 6358, 1912 and 1711 members; we do not print them, so at that size the check is delegated to a program.

Theorem 2. $\pi(\#1395) = \pi(\#44170) = \pi(\#44172) = 14$ and $\pi(\#6698) = 12$. *In particular #6698 is Class 0, so J_3 is not the only Class 0 cubic 12-vertex graph of girth 3 and diameter 3.*

The upper bounds in Theorem 2 rest on exhaustive search, organised by the two elementary lemmas of §4; the lower bounds are certificates. Nothing in the source paper is contradicted: both of its proved statements, $\pi(J_3) = 12$ and $\pi \geq 13$ for the three graphs, are confirmed here.

2. THE GRAPHS

Vertices are $0, \dots, 11$; the lists below are the adjacency lists served by the House of Graphs REST API for the four ids, under API index = HoG label – 1. Every statement in this note about the five named graphs is a statement about these lists as printed, so a reader who distrusts the database can still check it.

#1395 (the truncated tetrahedral graph):

```
0: 1, 2, 3   1: 0, 8, 9   2: 0, 3, 11  3: 0, 2, 10
4: 5, 6, 10  5: 4, 7, 11  6: 4, 8, 10  7: 5, 9, 11
8: 1, 6, 9   9: 1, 7, 8  10: 3, 4, 6  11: 2, 5, 7
```

#6698 (the graph the source asks about):

```
0: 1, 2, 3   1: 0, 8, 9   2: 0, 3, 11  3: 0, 2, 10
4: 5, 8, 10  5: 4, 8, 11  6: 7, 9, 11  7: 6, 9, 10
8: 1, 4, 5   9: 1, 6, 7  10: 3, 4, 7  11: 2, 5, 6
```

#44170:

```
0: 1, 2, 3   1: 0, 10, 11  2: 0, 10, 11  3: 0, 8, 9
4: 5, 6, 8   5: 4, 7, 8   6: 4, 9, 10   7: 5, 9, 11
8: 3, 4, 5   9: 3, 6, 7   10: 1, 2, 6   11: 1, 2, 7
```

#44172:

```
0: 1, 2, 3   1: 0, 9, 11  2: 0, 10, 11  3: 0, 7, 8
4: 5, 6, 8   5: 4, 9, 10  6: 4, 9, 10   7: 3, 8, 11
8: 3, 4, 7   9: 1, 5, 6  10: 2, 5, 6   11: 1, 2, 7
```

The control J_3 is taken from the source's own figure for J_3 with its labels $A, \dots, L$ read as $0, \dots, 11$, i.e. the edges $AB, AC, AD, BE, BF, CG, CH, DI, DJ, EK, FL, EF, GK, IK, HL, JL, HI, GJ$.

Proposition 3. *Each of the five graphs is a simple connected cubic graph on 12 vertices with 18 edges, girth 3, radius 3 and diameter 3; they are pairwise non-isomorphic. Moreover #1395 is isomorphic to the truncation of K_4 , and the graph just read off the source's figure is isomorphic to the standard Flower snark J_3 (three stars $K_{1,3}$ with centres a_i and leaves b_i, c_i, d_i ; a triangle on the b_i ; a 6-cycle $c_0c_1c_2d_0d_1d_2$).*

Every clause of Proposition 3 is recomputed from the printed lists by the accompanying program.

3. THREE CERTIFICATES

For a configuration C let $\text{Reach}(C)$ be the set of configurations obtainable from C by (possibly empty) sequences of pebbling moves.

Lemma 4. *Let $r \in V(G)$ and let S be a set of configurations with $C \in S$, S closed under pebbling moves, and $D(r) = 0$ for every $D \in S$. Then C is r -unsolvable, and $\pi(G) \geq |C| + 1$ where $|C|$ is the total of C .*

Proof. By induction along a sequence of moves, every configuration reachable from C lies in S , hence has no pebble on r ; so no sequence of moves places a pebble on r . As C has total $|C|$ and is not r -solvable, $\pi(G) > |C|$. $\square$

The set $S = \text{Reach}(C)$ is closed by construction, and it is finite because a pebbling move strictly decreases the total. So a certificate is just the finite list $\text{Reach}(C)$, and checking it is bookkeeping: generate the moves, confirm none lands a pebble on r .

TABLE 1. The three witnesses. Each has total 13 and $C(r) = 0$, and each $\text{Reach}(C)$ is closed under moves and free of pebbles on r .

	graph	r	$C = (C(0), \dots, C(11))$	$ \text{Reach}(C) $
W_1	#1395	0	(0, 0, 0, 0, 3, 4, 1, 3, 0, 1, 1, 0)	6358
W_2	#44170	1	(0, 0, 1, 1, 5, 5, 0, 0, 1, 0, 0, 0)	1912
W_3	#44172	5	(3, 1, 1, 0, 0, 0, 1, 7, 0, 0, 0, 0)	1711

Proof of Theorem 1. Apply Lemma 4 to $S = \text{Reach}(W_i)$ for $i = 1, 2, 3$ with the targets of Table 1. Each W_i has total 13, so $\pi \geq 14$ for the corresponding graph. $\square$

The source's own three 12-pebble configurations, read off its figure of unsolvable configurations under API index = label - 1, are re-certified unsolvable in the same way, which confirms its bound $\pi \geq 13$: #1395, $r = 0$, (0, 0, 0, 0, 0, 7, 3, 0, 0, 1, 1, 0), $|\text{Reach}| = 1262$; #44170, $r = 5$, (5, 1, 1, 0, 0, 0, 0, 0, 0, 5, 0), 1251; and #44172, $r = 0$, (0, 0, 0, 0, 7, 1, 1, 0, 0, 1, 1, 1), 483.

4. THE EXACT VALUES

Fix G and a target r . A configuration with $C(r) \geq 1$ is r -solvable trivially, so let

$$U_s(r) = \{ C : \sum_v C(v) = s, C(r) = 0, C \text{ is } r\text{-unsolvable} \}.$$

Then $\pi(G) = \max_r \min\{s : U_s(r) = \emptyset\}$ follows from the definition together with Lemma 5 below. Two lemmas make $U_s(r)$ computable without ever enumerating the whole space.

Lemma 5 (downward closure). *If $C' \leq C$ pointwise and C is r -unsolvable, then C' is r -unsolvable. Consequently $U_s(r) = \emptyset$ implies $U_{s'}(r) = \emptyset$ for all $s' \geq s$, and every $C \in U_s(r)$ arises as $C' + e_v$ for some $C' \in U_{s-1}(r)$ and some $v \neq r$.*

Proof. Every move legal at C' is legal at C , and applying the same move to both preserves $C' \leq C$; so a sequence of moves solving C' solves C , and the first claim is the contrapositive. If $C \in U_{s+1}(r)$, remove one pebble from any $v \neq r$ with $C(v) \geq 1$: the result lies in $U_s(r)$ by the first claim, which gives both remaining assertions. $\square$

Lemma 6 (one-move recurrence). *Let C have total $s \geq 1$ and $C(r) = 0$. Then $C \in U_s(r)$ if and only if for every v with $C(v) \geq 2$: $r \notin N(v)$, and $C - 2e_v + e_u \in U_{s-1}(r)$ for every $u \in N(v)$.*

Proof. C is r -solvable iff some first move leads to an r -solvable configuration or places a pebble on r directly; a move from v to r is available exactly when $C(v) \geq 2$ and $r \in N(v)$. A move lowers the total by 1, so the recurrence is acyclic and needs no fixed point. Configurations all of whose entries are at most 1 admit no move and are therefore in $U_s(r)$, which is the base case. $\square$

Lemmas 5 and 6 give a complete algorithm: compute $U_0(r) = \{0\}$, and obtain $U_s(r)$ by adding one pebble in every possible way to every member of $U_{s-1}(r)$ — complete by Lemma 5 — and testing each candidate by Lemma 6. The search stops at the first empty level, which by Lemma 5 is $\min\{s : U_s(r) = \emptyset\}$.

Proof of Theorem 2. Running the algorithm on the printed adjacency lists for all 12 targets of each graph gives the level sizes $|U_s(r)|$ recorded in Table 2: for #6698 (and for the control J_3) every target already has $U_{12}(r) = \emptyset$, while $U_{11}(r) \neq \emptyset$, so $\pi(\#6698) = 12 = n$ and #6698 is Class 0; for the other three graphs $U_{13}(r) \neq \emptyset$ for at least one target and $U_{14}(r) = \emptyset$ for all targets, so $\pi = 14$. $\square$

The sweep accounts for the whole space: for a fixed target the configurations with $C(r) = 0$ and total s number $\binom{s+10}{10}$, which is 646 646 at $s = 12$ and 1 144 066 at $s = 13$, so the Class-0 statement for #6698 is a statement about all $12 \cdot 646\,646 = 7\,759\,752$ of them. The three witnesses of Table 1 are members of the corresponding U_{13} , which ties the certificates of §3 to the sweep.

So the answer to the source's restatement of its own question — “determine whether J_3 is the only Class 0 graph among the five with the same parameters” — is no: the graph printed here as #6698 is Class 0 as well, and it is not isomorphic to J_3 .

5. SCOPE

The upper bounds have no hand-checkable certificate here: that $U_{12} = \emptyset$ for #6698, and $U_{14} = \emptyset$ for the other three, is proved by exhaustion over

TABLE 2. $|U_s(r)|$ by target $r = 0, \dots, 11$, and the sums over all 12 targets. Every entry is a count of r -unsolvable configurations; a configuration with a pebble on r is r -solvable by definition and is not counted.

s	per-target vector	sum
12	#1395: (165, 165, 165, 165, 165, 165, 165, 165, 165, 165, 165, 165)	1980
12	#44170: (0, 229, 229, 0, 91, 91, 0, 0, 91, 3, 0, 0)	734
12	#44172: (8, 0, 0, 94, 4, 153, 153, 94, 18, 5, 5, 8)	542
12	#6698: (0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)	0
12	J_3 : (0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)	0
13	#1395: (2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2)	24
13	#44170: (0, 20, 20, 0, 4, 4, 0, 0, 4, 0, 0, 0)	52
13	#44172: (0, 0, 0, 0, 0, 6, 6, 0, 0, 0, 0, 0)	12
13	#6698: (0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)	0
14	all five graphs: (0, $\dots$, 0)	0

a search space whose completeness is itself proved (Lemma 5), but it is exhaustion, and no weight-function or LP-dual certificate for $\pi(\#6698) \leq 12$, checkable with integer arithmetic and no program, is offered. Theorem 1 asks for less trust, resting on the closure of three single configurations of a few thousand members each rather than on an exhaustive sweep, but those closures are not printed here either, so it too is a machine check, only a small and local one.

Only the five graphs printed in §2 are treated. How many cubic 12-vertex graphs of girth 3 and diameter 3 there are, and what their pebbling numbers are, is not settled here; in particular the source's report that the database holds five such graphs is neither used nor contradicted.

We give exact values, not a characterisation: nothing here explains why #6698 is Class 0 while #1395, #44170 and #44172 are not. Neighbouring cells are untouched: cubic 12-vertex graphs of girth 4 or girth 5 at diameter 3, girth-3 members of diameter at least 4, every $n \neq 12$, and the source's own question about extending its method beyond diameter 3. The best published diameter-3 upper bound, $\pi \leq \frac{3}{2}n + O(1)$ [6, 7], gives roughly 20 at $n = 12$ and cannot separate 12 from 13 from 14, so no prior theorem forces any of the values above.

Remark (verification). The program `verify.py` accompanying this note recomputes, from the adjacency lists of §2 transcribed into it, using only the Python standard library and exact integer arithmetic, the graph parameters and identifications of Proposition 3, the reachability certificates of §3 with their sizes, all of Table 2, and the pebbling number of each of the five named graphs; its recorded run also reports quantities this note does not print, among them the pebbling numbers of eleven further cubic 12-vertex

graphs of girth 3 and diameter 3, which nothing here claims. It does not parse this note, so the transcription of the printed objects into its inputs is not checked by it; it does not fetch or parse any House of Graphs record, so the identification of the four adjacency lists with the ids #1395, #6698, #44170 and #44172 is not checked by it either, as its own scope note records.

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